

Rich User Context for Human-AI Interaction

AI systems are increasingly expected to adapt to the people they serve. Yet meaningful personalization demands that users articulate their preferences, goals, and constraints in detail, repeatedly. This is costly and often impossible: people know a response missed the mark without being able to say why. Professionals whose work is understanding people (counselors, clinicians, coaches) face this problem in sharper form: their time with each person is scarce, yet they must personalize to every new individual. In both cases the burden of making implicit context usable falls on people, and systems fall back on coarse proxies (demographics, click histories).

My research builds AI agents that *share this burden*, acquiring rich context about a person with as little friction as possible and judged by long-run benefit rather than a single exchange. Without interaction, I *synthesize* high-value signals about a person (Pillar 1); with it, I train agents to *elicit* only what matters (Pillar 2); and because inferring about people carries risk, I evaluate personalization and safety over long horizons (Pillar 3).

Pillar 1: Synthesizing rich user signals when interaction is unavailable. Much of what makes personalization work is in principle verbalizable (a person’s experiences, values, reasoning) but rarely stated, and behavioral logs are a poor substitute. I have argued that user *rationales*, short explanations of why someone holds a preference, are the missing signal: in PrimeX [3], we collected opinions, explanations, and worldview measures (Primal World Beliefs) from 858 US residents and showed that both *substantially improve opinion* prediction over demographics. Since such explanations are rarely available in practice, our PB&J framework [10] *synthesizes* them: an LLM reasons about the experiences, personality traits, and beliefs behind a user’s judgment, and the resulting rationale-augmented personas *outperform* demographic- and history-based personas on public-opinion and movie-preference prediction tasks, approaching human-written rationales. Most recently, PALMs [1] scales this to populations: rationales grounded in five construct families (personality, cultural dimensions, values, moral foundations, primal beliefs) serve as latent supervision during preference optimization, yielding models aligned to five countries that improve 8.6% over the strongest baseline and transfer to OOD tasks like personalized reward modeling. Grounding synthesis in psychological theory also makes these signals *inspectable*, so stereotyped inferences can be caught rather than silently acted on; for a professional meeting someone new, they offer a starting hypothesis in their field’s own vocabulary. **Future vision.** These representations are static snapshots; I want to make them *longitudinal*, updating a user model as needs shift and learning which signals stay predictive over weeks, not a single session.

Pillar 2: Learning to ask: proactive, low-friction elicitation. When interaction is possible, the lowest-friction system asks only for what it needs. My ongoing work [8] formulates this as multi-turn reinforcement learning: an assistant converses with a user, decides what to ask and when to stop, and is rewarded on the downstream decision (e.g., recommending an exercise routine given hidden constraints like injuries, or reaching a diagnosis based on clinical attributes). Training uses simulated users with controlled verbosity and willingness to share, who do not volunteer information. A key finding is that sparse task reward alone produces poor questioning: agents repeat themselves, bundle many requests into one turn, and ask what users cannot answer. We address this with dense, turn-level supervision from a *privileged* teacher that sees the hidden profile and critiques each question; distilling this into the policy improves task reward over RL methods like GRPO on both tasks while yielding questions that are significantly more answerable, less dense, and less repetitive. **Future vision.** First, objectives that reward a person’s *trajectory* across repeated sessions rather than per-episode task completion, under explicit budgets on user burden. Second, elicitation in service of professionals whose time is scarce: agents that gather routine context beforehand so a consultation goes to what needs a human, and, since answerability, density, and repetition are exactly what a supervisor corrects in a trainee’s questions, turning our judges of them into feedback for practitioners learning to ask [5].

Pillar 3: Evaluating personalization over the long horizon. Shifting inference onto the system risks misreading or stereotyping users [2, 6], and standard metrics largely miss this. In ELI-Why [7], we showed that LLM explanations tailored to different audiences match their intended audience only about half the time and are rated less useful; automatic rationale-quality judgments likewise diverge from human utility [9]. Such studies are cross-sectional; I am part of two ongoing efforts on the longitudinal failures I believe matter more. In a multi-institutional study, we log participants’ interactions with an assistant over 8+ weeks while systematically varying the information used to personalize for them, and have them produce real artifacts (images, websites) weekly, directly testing whether personalization compounds, plateaus, or misleads over time. In parallel, we are studying long-horizon safety: whether models that respond safely to single-turn mental-health queries still do so as a conversation unfolds over many turns, given evidence that LLMs degrade in extended interactions [4]. **Future vision.** I aim to turn these into reusable longitudinal evaluation frameworks and auditing tools that flag when a system’s model of a person diverges from their own or a professional’s assessment: prerequisites for safe deployment in settings like health.

Toward AI for long-term well-being. Health and well-being, where professionals must understand each person from few, costly interactions, is where these pillars fit together most naturally, and where I want to take this work.

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